

TASK No 02

1. Implement Software based firewall and connect it with wazuh.
2. Configure and create rule on firewall and test it with home lab
 - BLOCK specific countries (example china Russia etc.) traffic
 - Create rules which restrict the user from specified websites
 - Administrator privileges rule
3. Try to Monitor the files and logs on wazuh
4. Create reports put analysis on the report regarding the traffic you have observed.

Integration of pfSense with wazuh:

Introduction

Integration of pfsense with wazuh to enhance network security by connected with wazuh, wazuh is open source security platform ability to security monitoring with pfsense firewall and route it. This provide the real-time log monitoring and detection of threat via wazuh dashboard.

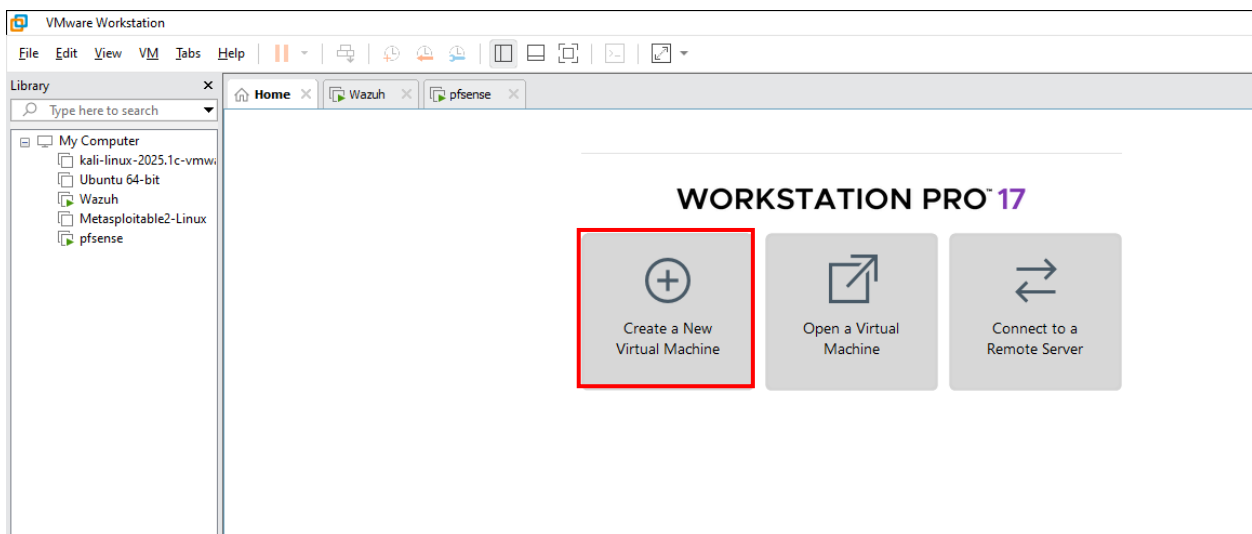
Requirments

- Vmware Workstation pro
 - Wazuh & wazuh-agent
 - Kali Linux
 - Pfsense (Firewall)
 - PfBlockerNG package
 - Squid
 - SquidGuard
1. **Implement Software based firewall and connect it with wazuh.**
 1. Download and Install VMware Workstation Pro Download Link (<https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>)

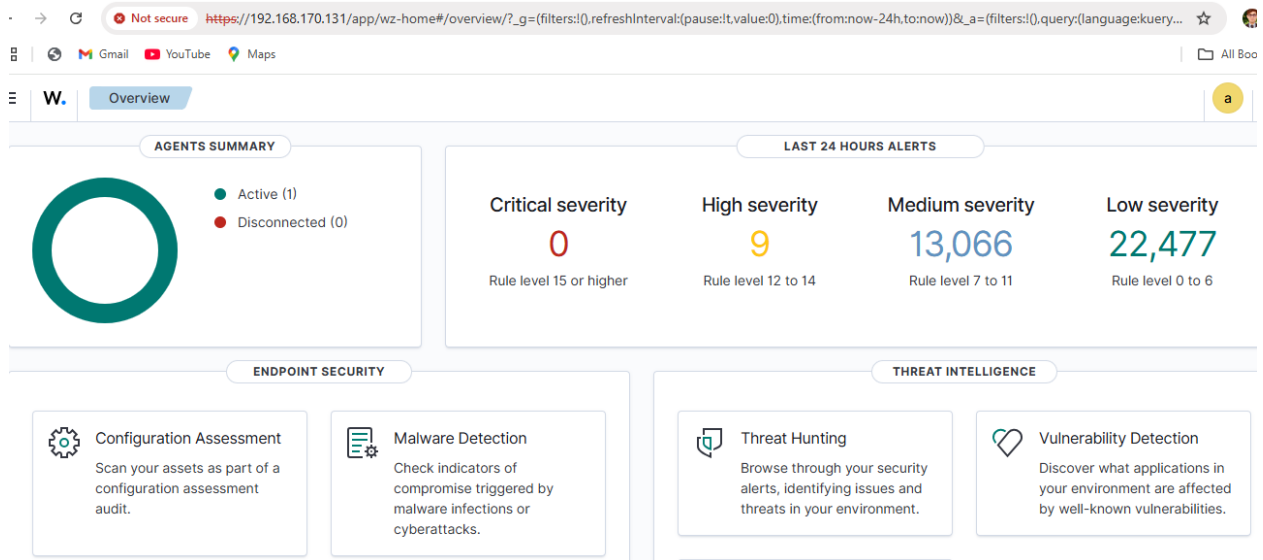
Fusion and Workstation

Run Windows, Linux and other virtual machines with VMware Workstation Pro for Windows and Linux or VMware Fusion for Mac, the industry standard desktop hypervisors.

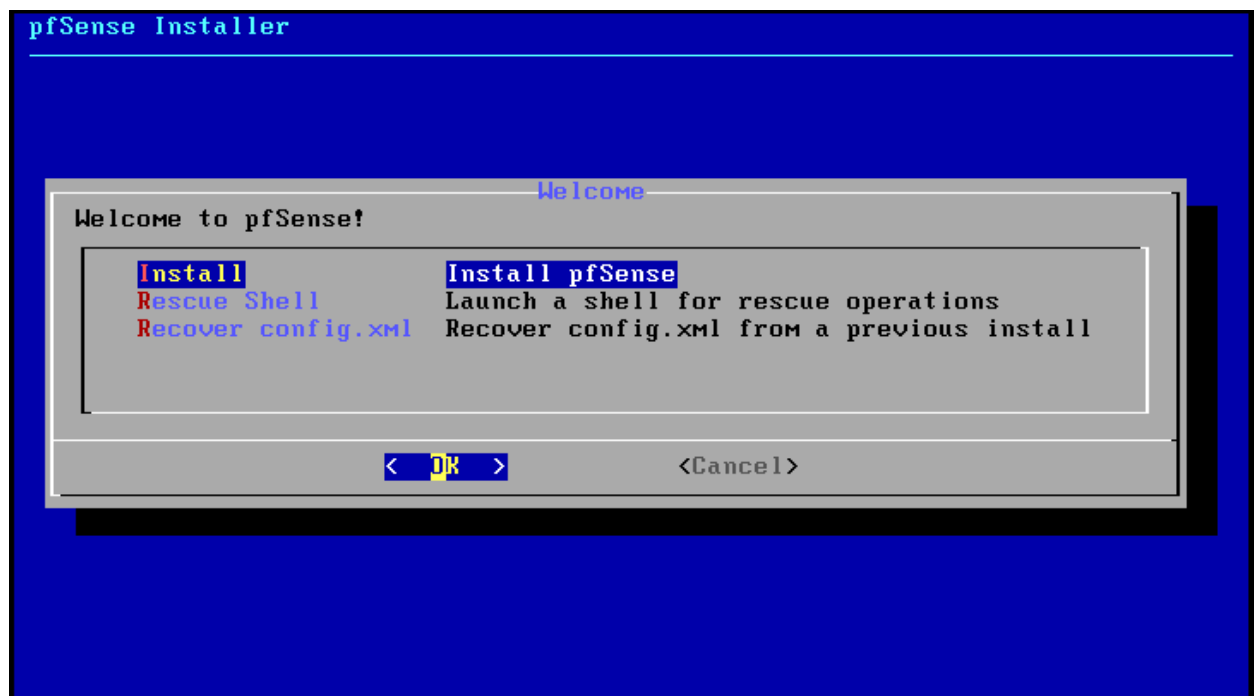
DOWNLOAD FUSION OR WORKSTATION



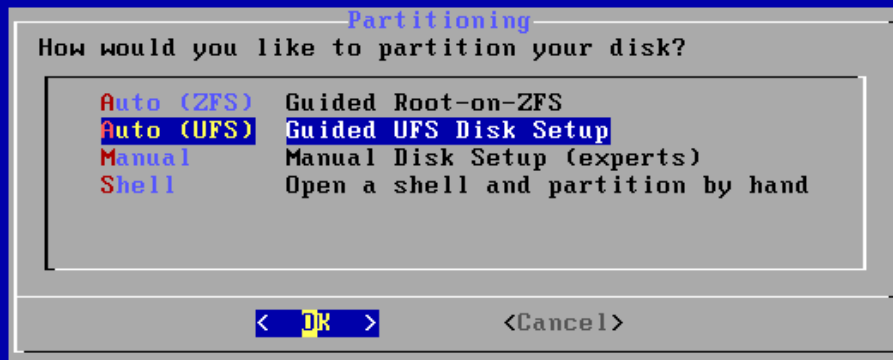
2. Download and Install Wazuh (OVA) File from (<https://documentation.wazuh.com/current/deployment-options/virtual-machine/virtual-machine.html>)



4. Download and install pfSense iso Image and create a New Virtual Machine in Vmware follow step by step instruction below:

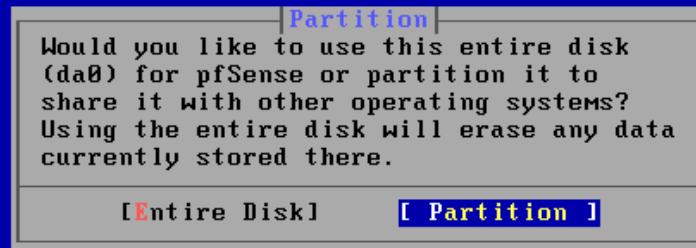


pfSense Installer

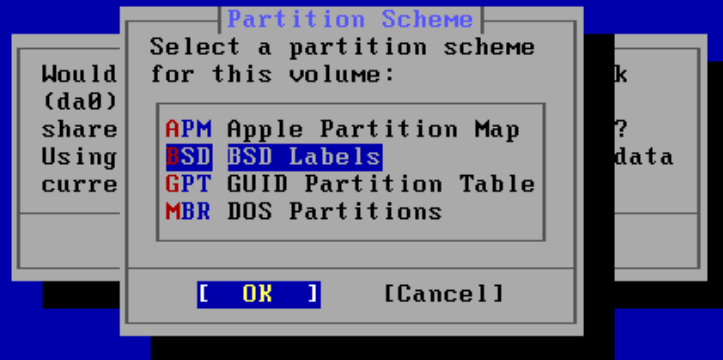


Menu options help choose which disk to setup using UFS and standard partitions

FreeBSD Installer

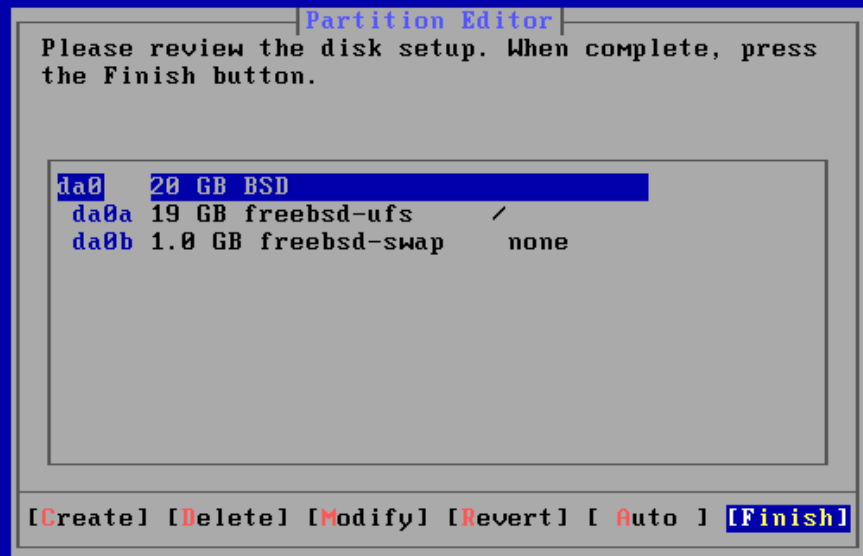


FreeBSD Installer



Bootable on most x86 systems

FreeBSD Installer



FreeBSD Installer

Partition Editor

Please review the disk setup. When complete, press the Finish button.

Confirmation

Your changes will now be written to disk. If you have chosen to overwrite existing data, it will be PERMANENTLY ERASED. Are you sure you want to commit your changes?

[**Commit**] [Revert & Exit] [**Back**]

[Create] [Delete] [Modify] [Revert] [**Auto**] [**Finish**]

pfSense Installer

Archive Extraction

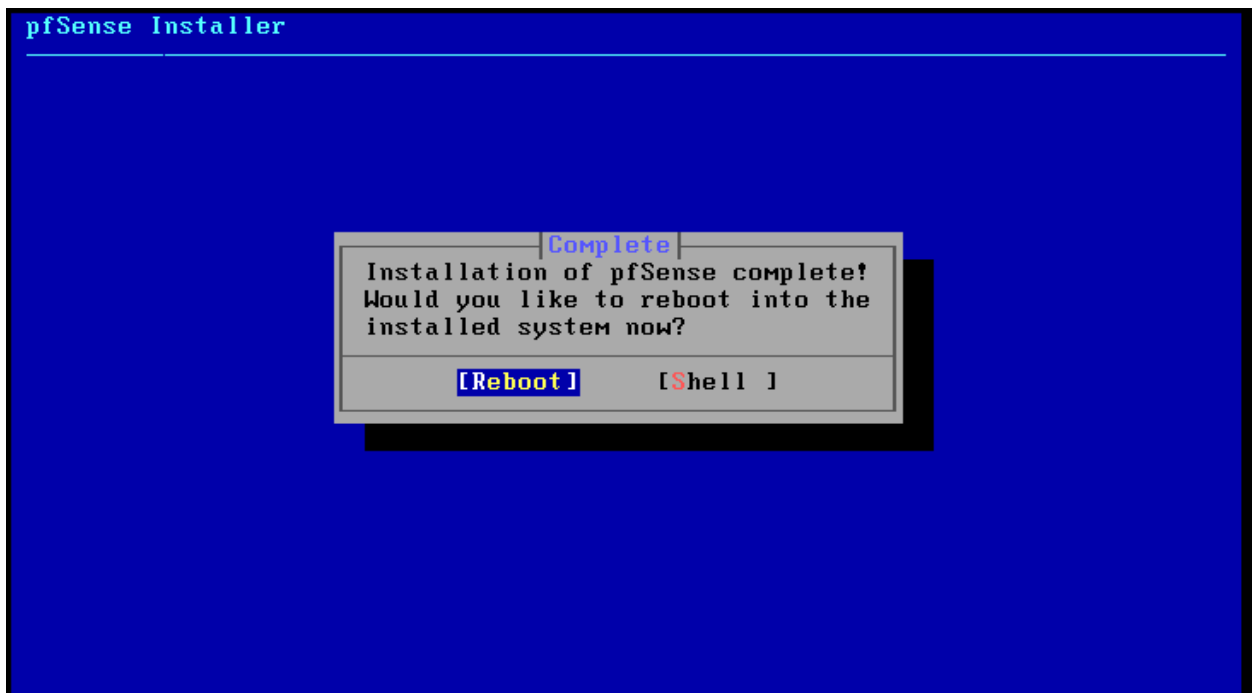
base.txz [ 9%]

Extracting distribution files...

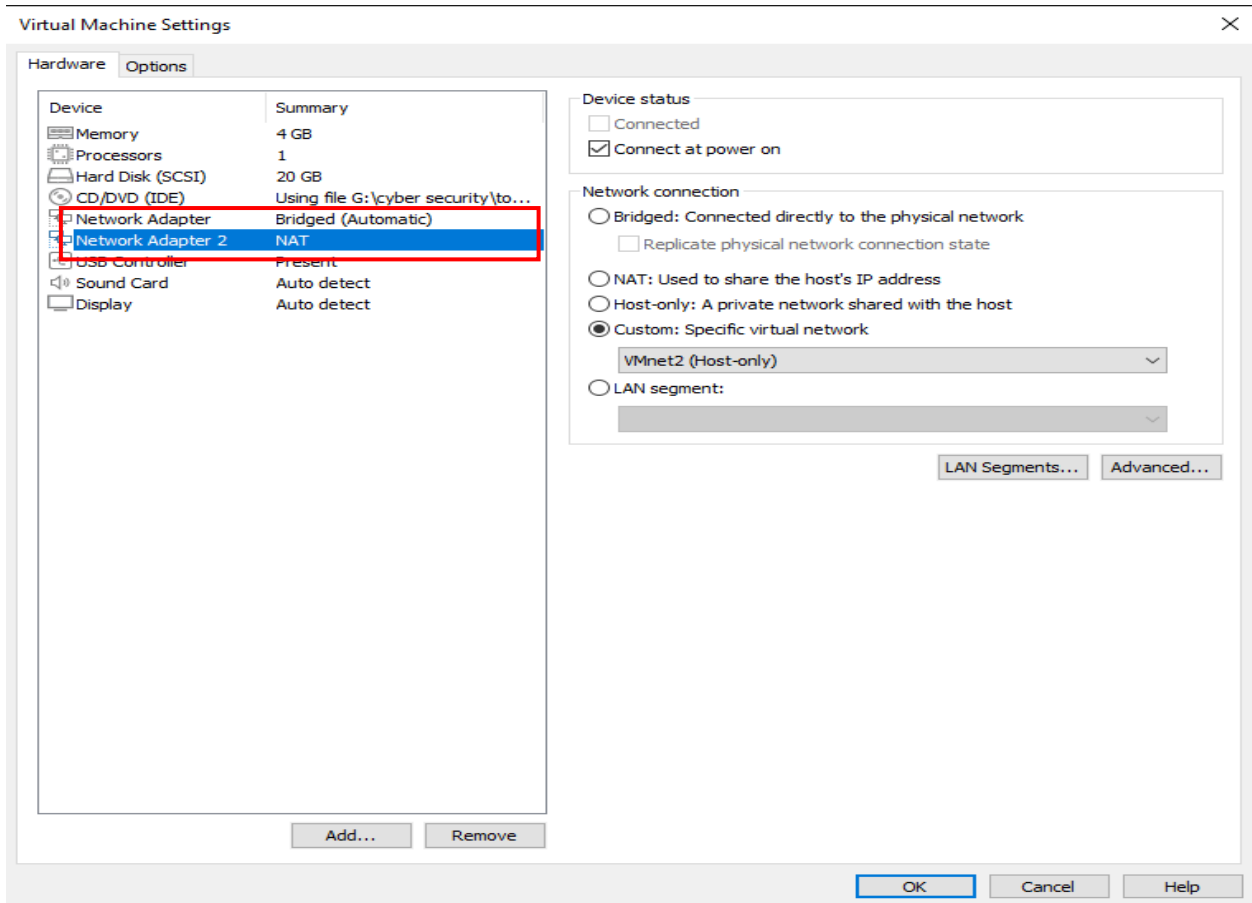
Overall Progress

 9%

4461 files read @ 743.5 files/sec.



Set two Network Adaptor one is Bridged and other is custom VMnet2 (Host-only)



Set IP Interface setup instruction in the Below Screenshot:

```
0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell

Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2

Configure IPv4 address LAN interface via DHCP? (y/n) n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.2.10
```

```
Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) n

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) n
Disabling IPv4 DHCPD...
Disabling IPv6 DHCPD...

Do you want to revert to HTTP as the webConfigurator protocol? (y/n) y
```

```
Do you want to revert to HTTP as the webConfigurator protocol? (y/n) y

Please wait while the changes are saved to LAN...
Reloading filter...
Reloading routing configuration...
DHCPD...
Restarting webConfigurator...

The IPv4 LAN address has been set to 192.168.2.10/24
You can now access the webConfigurator by opening the following URL in your web
browser:
    http://192.168.2.10/

Press <ENTER> to continue.
```

Run the IP (192.168.2.10) in browser: by default username is "admin" and password is "pfsense"



SIGN IN

Username

Password

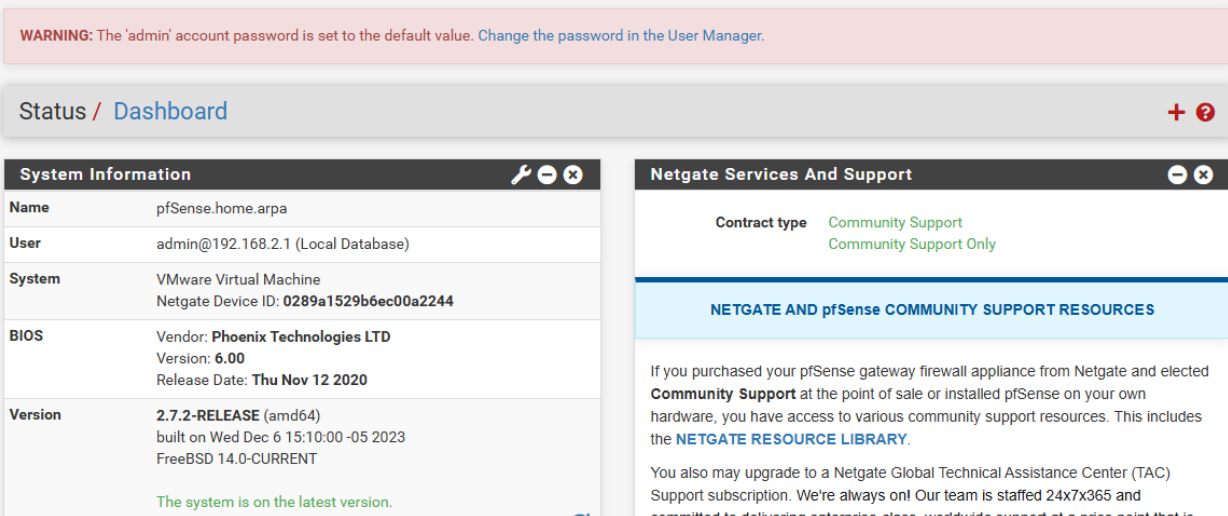
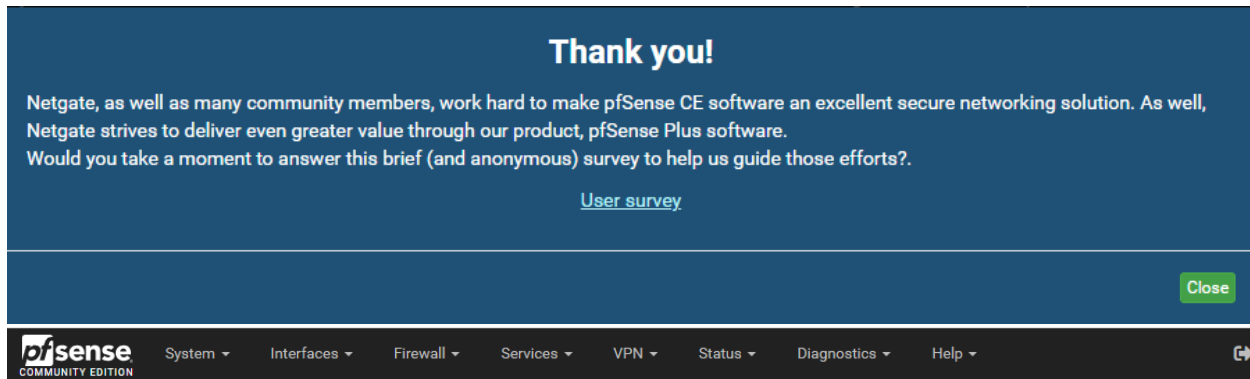
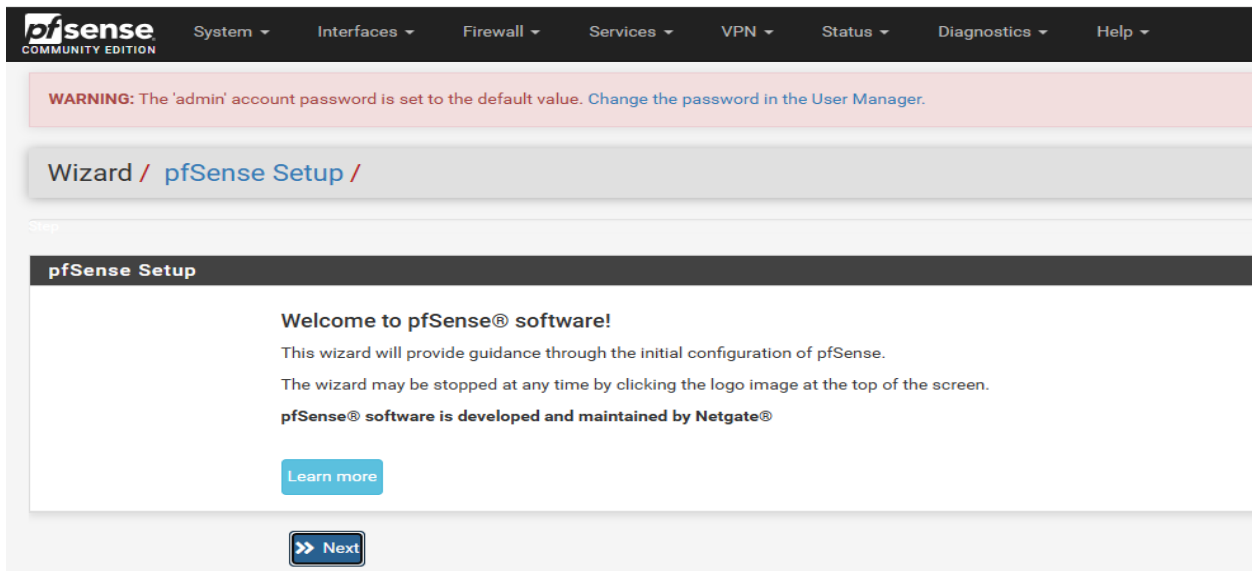
Follow all of the instruction and set DNS Server primary as 8.8.8.8 and secondary is 8.8.4.4

Primary DNS Server

Secondary DNS Server

Override DNS ☒

Allow DNS servers to be overridden by DHCP/PPP on WAN



5. Enable Remote Logging in pfSense to Wazuh go to status -> syslog.

Wazuh IP (192.168.170.131)

pfSense
COMMUNITY EDITION

System

Interfaces

Firewall

Services

VPN

Status

Diagnostics

Help

WARNING: The 'admin' account password is set to the default value. [Change the password in the Settings page.](#)

Status / Dashboard

System Information

Name

pfSense.home.arpa

User

admin@192.168.2.1 (Local Database)

System

VMware Virtual Machine
Netgate Device ID: 0289a1529b6ec00a2244

BIOS

Vendor: Phoenix Technologies LTD
Version: 6.00
Release Date: Thu Nov 12 2020

Version

2.7.2-RELEASE (amd64)
built on Wed Dec 6 15:10:00 -05 2023
FreeBSD 14.0-CURRENT

The system is on the latest version.

Version information updated at Mon Jul 7 11:50:46 -05 2025

CPU Type

Intel(R) Core(TM) i3-3217U CPU @ 1.80GHz
AES-NI CPU Crypto: No

Captive Portal

CARP (failover)

Dashboard

DHCP Leases

DHCPv6 Leases

DNS Resolver

Filter Reload

Gateways

Interfaces

IPsec

Monitoring

NTP

OpenVPN

Queues

Services

System Logs

Traffic Graph

UPnP & NAT-PMP

Status / System Logs / Settings

?

System

Firewall

DHCP

Authentication

IPsec

PPP

PPPoE/L2TP Server

OpenVPN

NTP

Packages

Settings

General Logging Options

Log Message Format

BSD (RFC 3164, default)

The format of syslog messages written to disk locally and sent to remote syslog servers (if enabled).
Changing this value will only affect new log messages.

Forward/Reverse Display

☐ Show log entries in reverse order (newest entries on top)

GUI Log Entries

500

This is only the number of log entries displayed in the GUI. It does not affect how many entries are contained in the actual log files.

Log firewall default blocks

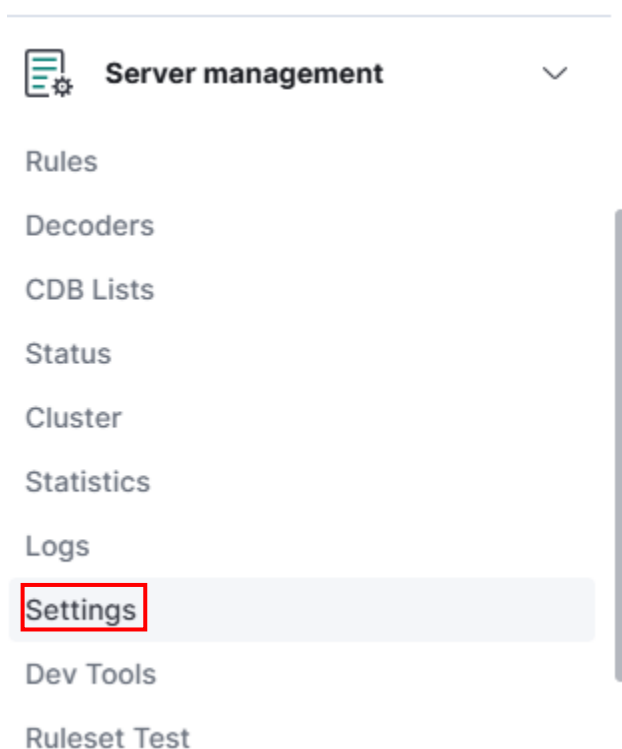
☒ Log packets matched from the default block rules in the ruleset
Log packets that are **blocked** by the implicit default block rule. - Per-rule logging options are still respected.

☐ Log packets matched from the default pass rules put in the ruleset
Log packets that are **allowed** by the implicit default pass rule. - Per-rule logging options are still respected.

☒ Log packets blocked by 'Block Bogon Networks' rules

Remote Logging Options		
Enable Remote Logging	☒ Send log messages to remote syslog server	
Source Address	LAN This option will allow the logging daemon to bind to a single IP address, rather than all IP addresses. If a single IP is picked, remote syslog servers must all be of that IP type. To mix IPv4 and IPv6 remote syslog servers, bind to all interfaces. NOTE: If an IP address cannot be located on the chosen interface, the daemon will bind to all addresses.	
IP Protocol	IPv4 This option is only used when a non-default address is chosen as the source above. This option only expresses a preference; If an IP address of the selected type is not found on the chosen interface, the other type will be tried.	
Remote log servers	192.168.170.131:514	IP[:port]
Remote Syslog Contents	☒ Everything ☐ System Events ☐ Firewall Events ☐ DNS Events (Resolver/unbound, Forwarder/dnsmasq, filterdns) ☐ DHCP Events (DHCP Daemon, DHCP Relay, DHCP Client) ☐ PPP Events (PPPoE WAN Client, L2TP WAN Client, PPTP WAN Client) ☐ General Authentication Events	

- Custom Syslog configuration click on three line in the wazuh left side corner then go to server management then click on settings in the below screenshot:



Click on Edit Configuration for edit remote connection

Configuration		Refresh Edit configuration ?
Main configurations		
Name	Description	
Global Configuration	Global and remote settings	
Cluster	Master node configuration	
Registration Service	Automatic agent registration service	

Edit pfsense Firewall with wazuh connection then save and Restart wazuh-manager

Allow IP of pfsense 192.168.2.10 and local IP of wazuh 192.168.170.131

see in the below Screenshot:

[<](#) Manager configuration [Refresh](#) [Save](#) [Restart Manager](#)

Edit ossec.conf of Manager

```
35 <connection>secure</connection>
36 <port>1514</port>
37 <protocol>tcp</protocol>
38 <queue_size>131072</queue_size>
39 </remote>
40
41 <!-- PfSense Firewall with Wazuh -->
42 <remote>
43 <connection>syslog</connection>
44 <port>514</port>
45 <protocol>udp</protocol>
46 <allowed-ips>192.168.2.10</allowed-ips>
47 <local_ip>192.168.170.131</local_ip>
48 </remote>
49
```

< pfsenseDecoder.xml

Decoders Test Save

```
1 <!-- add new decoder for pfsense "Filename: pfsense-custom-decoder.xml"-->
2 <decoder name="pfsense-custom">
3   <prematch>filterlog</prematch>
4 </decoder>
5 <decoder name="pfsense-fields">
6   <parent>pfsense-custom</parent>
7   <regex>^(\\w+)[\\d+]: \\S*,\\S*,\\S*,(\\S*),\\S*,\\S*,(\\S*),\\S*,\\S*,\\S*,\\S*,\\S*,\\S*,(\\S*),\\S*,(\\S*),\\S*,(\\S*),\\S*,(\\S*),\\S*,(\\S*),\\S*</regex>
8   <order>logsourcesource,id,action,protocol,srcip,dstip,srcport,dstport</order>
9 </decoder>
```

 Decoders Test

```

1 <!-- add new decoder for pfsense "Filename: pfsense-custom-decoder.xml"-->
2 <decoder name="pfsense-custom">
3   <prematch>filterlog</prematch>
4 </decoder>
5 <decoder name="pfsense-fields">
6   <parent>pfsense-custom</parent>
7   <regex>^(w+)[\d\ ]: \$*,\$*,\$*,(\$*),\$*,\$*,(\$*),\$*,\$*,\$*,\$*,\$*,\$*,\$*,(\$*),\$*,(\$*),(\$*),(\$*),(\$*),\$*</regex>
8   <order>logsource,id,action,protocol,srcip,dstip,srcport,dstport</order>
9 </decoder>

```

Decoders (5)

Manage decoders files

Add new decoders file

Refresh

Export formatted

From here you can manage your decoders.

relative_dirname=etc/decoders

WQL

Custom decoders

Name	Program name	Order	File ↑	Path
local_decoder_example	{"pattern":"local_decoder_example"}		local_decoder.xml	etc/decoders
pfsense-custom			pfsenseDecoder.xml	etc/decoders
pfsense-fields		logsource,id,action,protocol,srcip,dstip,srcport,dstport	pfsenseDecoder.xml	etc/decoders
squidguard-custom			SquidGuardDecoder.xml	etc/decoders
squidguard-fields		squidguard_action,srcip,dstip,url,reason	SquidGuardDecoder.xml	etc/decoders

Rows per page: 10

< 1 >

Create New custom Rules see in below screenshort:

 **Server management** 

Rules

Decoders

CDB Lists

Status

Cluster

Statistics

Logs

Settings

Dev Tools

Ruleset Test

Rules (0) [Manage rules files](#) [Add new rules file](#) [Refresh](#) [Export formatted](#)

From here you can manage your rules.

Search WQL Custom rules

ID ↑	Description	Groups	Regulatory compliance	Level	File	Path
No items found						

[Ruleset Test](#) [Save](#)

```

1 <!-- add new rule for pfsense "Filename: pfsense-rules"-->
2 <group name="pfsense-custom">
3   <rule id="100900" level="1">
4     <decoded_as>pfsense-custom</decoded_as>
5     <field name="logsource">filterlog</field>
6     <description>pfsense firewall rules grouped.</description>
7   </rule>
8
9   <rule id="100901" level="2">
10    <if_sid>100900</if_sid>
11    <action>pass</action>
12    <options>no_full_log</options>
13    <description>pfsense firewall allow event.</description>
14    <group>firewall_allow,pci_dss_1.4,gpg13_4.12,hipaa_164.312.a.1,nist_800_53_SC.7,tsc_CC6.7,tsc_CC6.8,</group>
15  </rule>
16
17   <rule id="100902" level="3">
18    <if_sid>100900</if_sid>
19    <action>block</action>
20    <options>no_full_log</options>

```

8. Add a custom logs to collect rules in wazuh to pfsense using decoders and rules to categorize events see in below screenshot:

884 hits

Jul 8, 2025 @ 00:18:33.627 - Jul 9, 2025 @ 00:18:33.627

Export Formatted 694 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	syscheck.path	syscheck.event	rule.des...	rule.level	rule.id
Jul 9, 2025 @ 00:18:17.171	win10	c:\program files (x86)\ossec-agent\queue\diff\file\cb91...	modified	Integrity ch...	7	550
Jul 9, 2025 @ 00:18:13.850	win10	c:\program files (x86)\ossec-agent\queue\diff\file\c6fb0...	modified	Integrity ch...	7	550
Jul 9, 2025 @ 00:18:10.940	win10	c:\program files (x86)\ossec-agent\queue\diff\file\c31c...	modified	Integrity ch...	7	550
Jul 9, 2025 @ 00:18:08.489	win10	c:\program files (x86)\ossec-agent\queue\diff\file\bde8...	modified	Integrity ch...	7	550
Jul 9, 2025 @ 00:18:04.207	win10	c:\program files (x86)\ossec-agent\queue\diff\file\b5a6...	modified	Integrity ch...	7	550
Jul 9, 2025 @ 00:18:03.378	win10	c:\program files (x86)\ossec-agent\queue\diff\file\b4b4...	modified	Integrity ch...	7	550
Jul 9, 2025 @ 00:17:56.827	win10	c:\program files (x86)\ossec-agent\queue\diff\file\a72e...	modified	Integrity ch...	7	550
Jul 9, 2025 @ 00:17:56.249	win10	c:\program files (x86)\ossec-agent\queue\diff\file\a64a...	modified	Integrity ch...	7	550
Jul 9, 2025 @ 00:17:52.691	win10	c:\program files (x86)\ossec-agent\queue\diff\file\a15c...	modified	Integrity ch...	7	550
Jul 9, 2025 @ 00:17:47.921	win10	c:\program files (x86)\ossec-agent\queue\diff\file\999b...	modified	Integrity ch...	7	550
Jul 9, 2025 @ 00:17:46.886	win10	c:\program files (x86)\ossec-agent\queue\diff\file\98ed...	modified	Integrity ch...	7	550
Jul 9, 2025 @ 00:17:46.542	win10	c:\program files (x86)\ossec-agent\queue\diff\file\9888...	modified	Integrity ch...	7	550

2. Configure and create rule on firewall and test it with home lab

a. BLOCK specific countries (example china Russia etc.) traffic

1. Install PfBlockerNG package goto system /package manager/ available packages, search PfBlockerNG this for Blocking GeoIP addresses.

System / Package Manager / Available Packages

Installed Packages Available Packages

Search

Search term PfBlockerNG Both Search Clear

Enter a search string or *nix regular expression to search package names and descriptions.

Packages

Name	Version	Description
pfBlockerNG-devel	3.2.0_20	pfBlockerNG-devel is the Next Generation of pfBlockerNG. Manage IPv4/v6 List Sources into 'Deny, Permit or Match' formats. GeoIP database by MaxMind Inc. (GeoLite2 Free version). De-Duplication, Suppression, and Reputation enhancements. Provision to download from diverse List formats. Advanced Integration for Proofpoint ET IQRisk IP Reputation Threat Sources. Domain Name (DNSBL) blocking via Unbound DNS Resolver.

+ Install

2. Goto Firewall click on PfBlockerNG configure PfBlockerNG

System Interfaces Firewall Services VPN Status Diagnostics

Aliases
NAT
pfBlockerNG
Rules
Schedules
Traffic Shaper
Virtual IPs

pfSense.home.arp
admin@192.168.2.1 (Local Database)
VMware Virtual Machine
Netgate Device ID: 0289a1529b6ec00a2244
Vendor: Phoenix Technologies LTD
Version: 6.00
Release Date: Thu Nov 12 2020

Netgate Services
Contract type
NETGATE AN
If you purchased your p
Community Support a

3. Open pfBlockerNG Goto General to enable pfBlockerNG

Firewall / pfBlockerNG

General
IP
DNSBL
Update
Reports
Feeds
Logs
Sync
Wizard

General Settings

Links
Firewall Aliases
Firewall Rules
Firewall Logs

pfBlockerNG
☒ Enable

Note: Context help is available on various pages by clicking the 'blue infoblock' icons -->

Keep Settings
☒ Enable

Note: With 'Keep settings' enabled, pfBlockerNG will maintain run state on Installation/Upgrade. If 'Keep Settings' is not 'enabled' on pkg Install/De-Install, all settings will be Wiped!

Note: To clear all downloaded lists, uncheck these two checkboxes and 'Save'. Re-check both boxes and run a 'Force Update|Reload'

CRON Settings

Every hour
00
0
0

Default: Every hour
Select the Cron hour interval.

Default: :00
Select the Cron update minute.

Default: 0
Select the Cron start hour.

Default: 0
Select the 'Daily/Weekly' start hour.

4. IP Configuration and Maxmind licences key and ID first enable De-Dublication and open Browser search maxmind website for licence key sign in and generate licence key and ID

Account ID:

Description	License key	Key created	Created by	Last used		
License Key #2	UE817K...	2025-07-08 07:02:07 UTC	mkhubabktk@gmail.com	2025-07-08 07:36:32 UTC		
License Key #1	N7WkX9...	2025-06-30 11:08:50 UTC	mkhubabktk@gmail.com	—		
License Key #0	5uaJTr...	2024-07-20 11:17:25 UTC	mkhubabktk@gmail.com	—		

Generate new license key

[General](#) [IP](#) [DNSBL](#) [Update](#) [Reports](#) [Feeds](#) [Logs](#) [Sync](#)[IPv4](#) [IPv6](#) [GeoIP](#) [Reputation](#)

IP Configuration

Links	Firewall Aliases Firewall Rules Firewall Logs
De-Duplication	☒ Enable Only used for IPv4 Deny Lists
CIDR Aggregation	☐ Enable Optimise CIDRs - merge contiguous CIDRs into larger CIDR blocks.
Suppression	☒ Enable Default enabled. This will prevent Selected IPs (and RFC1918/Loopback addresses) from being blocked. Only for IPv4 lists (/32 and /24). i
Force Global IP Logging	☐ Enable The global logging option is only used to force logging for all IP Aliases, and not to disable the logging of all IP Aliases. This overrides any logging settings in the GeoIP/IPv4/v6 tabs.

MaxMind GeoIP configuration

MaxMind Account ID	1040231 To utilize the free MaxMind GeoLite2 GeoIP functionality, you must first register for a free MaxMind user account. Visit the following Link to Register for a free MaxMind user account. Use the GeoIP Update version 3.1.1 or newer registration option.
MaxMind License Key	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 To utilize the free MaxMind GeoLite2 GeoIP functionality, you must first register for a free MaxMind user account. Visit the following Link to Register for a free MaxMind user account. Utilize the GeoIP Update version 3.1.1 or newer registration option.
MaxMind Localized Language	English Select the localized name data from the Language options available. Changes to the Locale will be executed in the background, and will take a few minutes to complete. Upon completion, a pfSense Notice will be generated.
MaxMind CSV Updates	☐ Check to disable MaxMind CSV updates This will disable the MaxMind monthly CSV GeoIP database cron update. This does not affect the MaxMind binary cron update that is used for other GeoIP functionality in the package.

5. Update the PfblockerNG take some time for updating

Firewall / pfBlockerNG / Update

General IP DNSBL **Update** Reports Feeds Logs Sync

Update Settings

Links Firewall Aliases Firewall Rules Firewall Logs

Status **Active pfBlockerNG CRON JOB** 
Refresh to update current status and time remaining.

Force Options **** AVOID **** Running these "Force" options - when CRON is expected to RUN!
Update: will process new changes and download new Alias/Lists.
Cron: will download any Alias/Lists that are within the Frequency Setting (due for Update).
Reload: will reload all Lists using the existing Downloaded files.
This is useful when Lists are out of "sync", Whitelisting, Blacklisting, Suppression, TLD or Reputation changes were made.

Select 'Force' option ☒ Update ☐ Cron ☐ Reload

Select 'Reload' option ☒ All ☐ IP ☐ DNSBL

Run **End View**

6. Open PfblockerNG goto IP and select GeoIP for blocking countries edit any of them do you want to block it, I want to block India, Afghanistan and Finland select them list Action option Deny Both (inbound, outbound) and Enabling Logging select disable save and update.

Firewall / pfBlockerNG / IP / GeoIP

General **IP** DNSBL Update Reports Feeds Logs Sync

IPv4 IPv6 **GeoIP** Reputation

GeoIP Summary

Name	Description	Action	Logging	
Top Spammers	GeoIP Top Spammers	Disabled	Enabled	
Africa	GeoIP Africa	Disabled	Enabled	
Antarctica	GeoIP Antarctica	Disabled	Enabled	
Asia	GeoIP Asia	Disabled	Enabled	
Europe	GeoIP Europe	Disabled	Enabled	
North America	GeoIP North America	Disabled	Enabled	
Oceania	GeoIP Oceania	Disabled	Enabled	
South America	GeoIP South America	Disabled	Enabled	

Use **CTRL+CLICK** to **select/unselect** the IPv4/6 Countries below as required.

AA ASIA UNDEFINED 6255147 (277)
AA ASIA UNDEFINED 6255147_rep (0)
Afghanistan [1149361] AF (215)
Afghanistan [1149361] AF_rep (1)
Armenia [174982] AM (327)
Armenia [174982] AM_rep (86)
Azerbaijan [587116] AZ (368)
Azerbaijan [587116] AZ_rep (41)
Bahrain [290291] BH (290)
Bahrain [290291] BH_rep (3)
Bangladesh [1210997] BD (2443)
Bangladesh [1210997] BD_rep (91)
Bhutan [1252634] BT (126)
Bhutan [1252634] BT_rep (2)
British Indian Ocean Territory [1282588] IO (38)
British Indian Ocean Territory [1282588] IO_rep (0)
Brunei [1820814] BN (128)
Brunei [1820814] BN_rep (4)
Cambodia [1831722] KH (480)
Cambodia [1831722] KH_rep (9)
China [1814991] CN (7957)
China [1814991] CN_rep (795)
Cocos (Keeling) Islands [1547376] CC (25)
Cocos (Keeling) Islands [1547376] CC_rep (0)
Georgia [614540] GE (467)
Georgia [614540] GE_rep (92)
Czechia [3077311] CZ_rep (1580)
Denmark [2623032] DK (3346)
Denmark [2623032] DK_rep (225)
Estonia [453733] EE (1044)
Estonia [453733] EE_rep (212)
Faroe Islands [2622320] FO (75)
Faroe Islands [2622320] FO_rep (0)
Finland [660013] FI (3063)
Finland [660013] FI_rep (202)
France [3017382] FR (32303)
France [3017382] FR_rep (4949)
Germany [2921044] DE (26688)
Germany [2921044] DE_rep (4336)
Gibraltar [2411586] GI (176)
Gibraltar [2411586] GI_rep (145)
Greece [390903] GR (1479)
Greece [390903] GR_rep (94)

AA ASIA UNDEFINED 6255147 (105)
Afghanistan [1149361] AF (589)
Armenia [174982] AM (425)
Azerbaijan [587116] AZ (258)
Bahrain [290291] BH (181)
Bangladesh [1210997] BD (1839)
Bhutan [1252634] BT (124)
British Indian Ocean Territory [1282588] IO (61)
Brunei [1820814] BN (88)
Cambodia [1831722] KH (217)
China [1814991] CN (11920)
Cocos (Keeling) Islands [1547376] CC (37)
Georgia [614540] GE (338)
Hong Kong [1810730] HK (4301)
India [1269750] IN (7115)
Indonesia [1643084] ID (4498)
Iran [130758] IR (679)
Iraq [99237] IQ (1141)
Israel [294640] IL (1424)
Japan [1861060] JP (5012)
Jordan [248816] JO (165)
Kazakhstan [1522867] KZ (756)
Kuwait [285570] KW (1007)
Kyrgyzstan [1527747] KG (151)
Laos [1655842] LA (117)
Lebanon [272103] LB (224)
Iceland [2629691] IS (843)
Ireland [2963597] IE (4529)
Isle of Man [3042225] IM (114)
Italy [3175395] IT (6760)
Jersey [3042142] JE (87)
Kosovo [831053] XK (16)
Latvia [458258] LV (436)
Liechtenstein [3042058] LI (430)
Lithuania [597427] LT (4003)
Luxembourg [2960313] LU (485)
Malta [2562770] MT (139)
Moldova [617790] MD (1269)
Monaco [2993457] MC (94)
Montenegro [3194884] ME (123)
North Macedonia [718075] MK (146)
Norway [3144096] NO (1915)
Poland [798544] PL (3066)

```
UPDATE PROCESS START [ v3.2.0_8 ] [ 07/8/25 05:21:10 ]

===[ DNSBL Process ]=====

Loading DNSBL Statistics... completed
Loading DNSBL SafeSearch... disabled
Loading DNSBL Whitelist... completed

[ StevenBlack_ADs ]          exists.

===[ GeoIP Process ]=====

[ pfB_Asia_v4 ]              exists.
[ pfB_Asia_v6 ]              Changes found... Updating

===[ IPv4 Process ]=====

[ Abuse_Feodo_C2_v4 ]        exists.
[ Abuse_SSLBL_v4 ]           exists.
[ CINS_army_v4 ]             exists.
[ ET_Block_v4 ]              exists.
[ ET_Comp_v4 ]               exists.
[ ISC_Block_v4 ]             exists.
[ Spamhaus_Drop_v4 ]         exists.
[ Spamhaus_eDrop_v4 ]        Downloading update .. 200 OK. completed ..
[ pfB_PRI1_v4 Spamhaus_eDrop_v4 ] No IPs found! Ensure only IP based Feeds are used! ]
```

8. Create firewall Alias and rules for blocking.

Firewall / Aliases / URLs

IP Ports URLs All

Firewall Aliases URLs			
Name	Type	Values	Description
countries_blocked	URL (IPs)		I want to some countries block on my site
pfB_Asia_v4	URL Table (IPs)	http://127.0.0.1:80/pfblockerng/pfblockerng.php? pfb=pfB_Asia_v4	pfBlockerNG Auto GeoIP Alias [AF,AF_rep,]
pfB_Asia_v6	URL Table (IPs)	http://127.0.0.1:80/pfblockerng/pfblockerng.php? pfb=pfB_Asia_v6	pfBlockerNG Auto GeoIP Alias [AF,AF_rep,IN,]
pfB_PRI1_v4	URL Table (IPs)	http://127.0.0.1:80/pfblockerng/pfblockerng.php? pfb=pfB_PRI1_v4	pfBlockerNG Auto Alias [Abuse_Feodo_C2_v4,Abuse_SSLBL_v4,CINS_army_v4,ET_Block_v4,ET_Comp_v4,ISC_Block_v4]

Add Firewall Rules in WAN

Firewall / Rules / WAN

Floating

WAN

LAN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	0/477 KiB	*	RFC 1918 networks	*	*	*	*	*		Block private networks	
☒	0/9 KiB	*	Reserved Not assigned by IANA	*	*	*	*	*		Block bogon networks	
☒	0/0 B	IPv4 *	pfB_Asia_v4	*	*	*	*	none		pfB_Asia_v4 auto rule	
☐	0/0 B	IPv6 *	pfB_Asia_v6	*	*	*	*	none		pfB_Asia_v6 auto rule	

↑ Add

↓ Add

🗑 Delete

🔄 Toggle

📄 Copy

💾 Save

+ Separator

9. Test in Home Lab

Check for logs to Block countries or not go to status select syslog /firewall check it to create firewall rules or not see in below screenshot:

Status / System Logs / Firewall / Normal View

System

Firewall

DHCP

Authentication

IPsec

PPP

PPPoE/L2TP Server

OpenVPN

NTP

Packages

Settings

Normal View

Dynamic View

Summary View

Last 500 Firewall Log Entries. (Maximum 500)

Action	Time	Interface	Rule	Source	Destination	Protocol
✗	Jul 8 05:53:24	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.84:137	192.168.100.255:137	UDP
✗	Jul 8 05:53:25	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.84:137	192.168.100.255:137	UDP
✗	Jul 8 05:53:26	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.84:137	192.168.100.255:137	UDP
✗	Jul 8 05:53:26	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.84:137	192.168.100.255:137	UDP
✗	Jul 8 05:53:27	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.84:137	192.168.100.255:137	UDP
✗	Jul 8 05:53:28	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.84:137	192.168.100.255:137	UDP
✗	Jul 8 05:53:29	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.84:137	192.168.100.255:137	UDP
✗	Jul 8 05:53:29	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.84:137	192.168.100.255:137	UDP
✗	Jul 8 05:53:30	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.84:137	192.168.100.255:137	UDP

Also check in wazuh manager for analysis to forward logs from pfsense see in below screenshot:

1,252 hits

Jul 8, 2025 @ 15:31:43.749 - Jul 8, 2025 @ 16:01:43.749

Export Formatted

749 available fields

Columns

Density

1 fields sorted

Full screen

	timestamp	agent.name	syscheck.path	syscheck.event	rule.des...	rule.level	rule.id
	Jul 8, 2025 @ 16:01:38.919	win10	c:\program files (x86)\ossec-agent\queue\diff\file\97e9...	modified	Integrity ch...	7	550
	Jul 8, 2025 @ 16:01:38.507	win10	c:\program files (x86)\ossec-agent\queue\diff\file\951...	modified	Integrity ch...	7	550
	Jul 8, 2025 @ 16:01:37.500	win10	c:\program files (x86)\ossec-agent\queue\diff\file\7a20...	modified	Integrity ch...	7	550
	Jul 8, 2025 @ 16:01:36.816	win10	c:\program files (x86)\ossec-agent\queue\diff\file\5e22...	modified	Integrity ch...	7	550
	Jul 8, 2025 @ 16:01:36.213	win10	c:\program files (x86)\ossec-agent\queue\diff\file\5c0f...	modified	Integrity ch...	7	550
	Jul 8, 2025 @ 16:01:35.651	win10	c:\program files (x86)\ossec-agent\queue\diff\file\460...	modified	Integrity ch...	7	550
	Jul 8, 2025 @ 16:01:35.089	win10	c:\program files (x86)\ossec-agent\queue\diff\file\315...	modified	Integrity ch...	7	550
	Jul 8, 2025 @ 16:01:33.854	win10	c:\program files (x86)\ossec-agent\queue\diff\file\effbd...	modified	Integrity ch...	7	550
	Jul 8, 2025 @ 16:01:32.059	win10	c:\program files (x86)\ossec-agent\queue\diff\file\ec31...	modified	Integrity ch...	7	550
	Jul 8, 2025 @ 16:01:31.029	win10	c:\program files (x86)\ossec-agent\queue\diff\file\ea25...	modified	Integrity ch...	7	550

3. Create Rules to Restrict Specific Websites

1. Install Squid and SquidGuard Packages

Go to System >Package Manager > Available Packages see in the below screenshot:

Installed Packages
Available Packages

Search

Search term

Both

Search

Clear

Enter a search string or *nix regular expression to search package names and descriptions.

Packages

Name	Version	Description	
Lightsquid	3.0.7_3	LightSquid is a high performance web proxy reporting tool. Includes proxy realtime statistics (SQStat). Requires Squid package.	
Package Dependencies:			
	0.4.46	High performance web proxy cache (3.5 branch). It combines Squid as a proxy server with its capabilities of acting as a HTTP / HTTPS reverse proxy. It includes an Exchange-Web-Access (OWA) Assistant, SSL filtering and antivirus integration via C-ICAP.	
Package Dependencies:			
	1.16.19	High performance web proxy URL filter.	
Package Dependencies:			

2. Configure Squid Proxy go to Services >Squid Proxy Server Enable Squid Proxy.

pfSense
COMMUNITY EDITION

System ▾ Interfaces ▾ Firewall ▾ **Services ▾** VPN ▾ Status ▾

WARNING: The 'admin' account password is set to the default value. Please change it immediately after installation.

System / Package Manager / Package Installation

pfSense-pkg-squidGuard installation successfully completed.

Installed Packages Available Packages **Package Installation**

Package Installation

Check documentation here:
<http://www.squidguard.org/Doc/>

To activate the changes do a `/usr/local/sbin/squid -k`
=====
Message from pfSense-pkg-squidGuard-1.16.19:

- Auto Config Backup
- Captive Portal
- DHCP Relay
- DHCP Server
- DHCPv6 Relay
- DHCPv6 Server
- DNS Forwarder
- DNS Resolver
- Dynamic DNS
- IGMP Proxy
- NTP
- PPPoE Server
- Router Advertisement
- SNMP
- Squid Proxy Server**
- Squid Reverse Proxy
- SquidGuard Proxy Filter
- UPnP & NAT-PMP

Squid General Settings

Enable Squid Proxy ☒ Check to enable the Squid proxy.
Important: If unchecked, ALL Squid services will be disabled and stopped.

Keep Settings/Data ☒ If enabled, the settings, logs, cache, AV defs and other data will be preserved across package reinstalls.
Important: If disabled, all settings and data will be wiped on package uninstall/reinstall/upgrade.

Listen IP Version
Select the IP version Squid will use to select addresses for accepting client connections.

3. Configure SquidGuard go to Services > SquidGuard Proxy Filter > General then Enable SquidGuard check it

Package / Proxy filter SquidGuard: General settings / General settings

General settings Common ACL Groups ACL Target categories Times Rewrites Blacklist Log XMLRPC Sync

General Options

Enable ☒ Check this option to enable squidGuard.

Important: Please set up at least one category on the 'Target Categories' tab before enabling. See [this link for details](#).
The Save button at the bottom of this page must be clicked to save configuration changes.
To activate squidGuard configuration changes, **the Apply button must be clicked.**



SquidGuard service state: **STOPPED**

4. Add Blacklist go to Common ACL add Target Rules List to block Restricted sites I have Block Facebook and Tiktok on my site:

Proxy filter SquidGuard: Target categories / Edit / Target categories

General settings Common ACL Groups ACL Target categories Times Rewrites Blacklist Log XMLRPC Sync

General Options

Name
Enter a unique name of this rule here.
The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.

Order
Select the new position for this target category. Target categories are listed in this order on ACLs and are matched from the top down in sequence.

Domain List

General Options

Target Rules

Target Rules List  

ACCESS: 'whitelist' - always pass; 'deny' - block; 'allow' - pass, if not blocked.

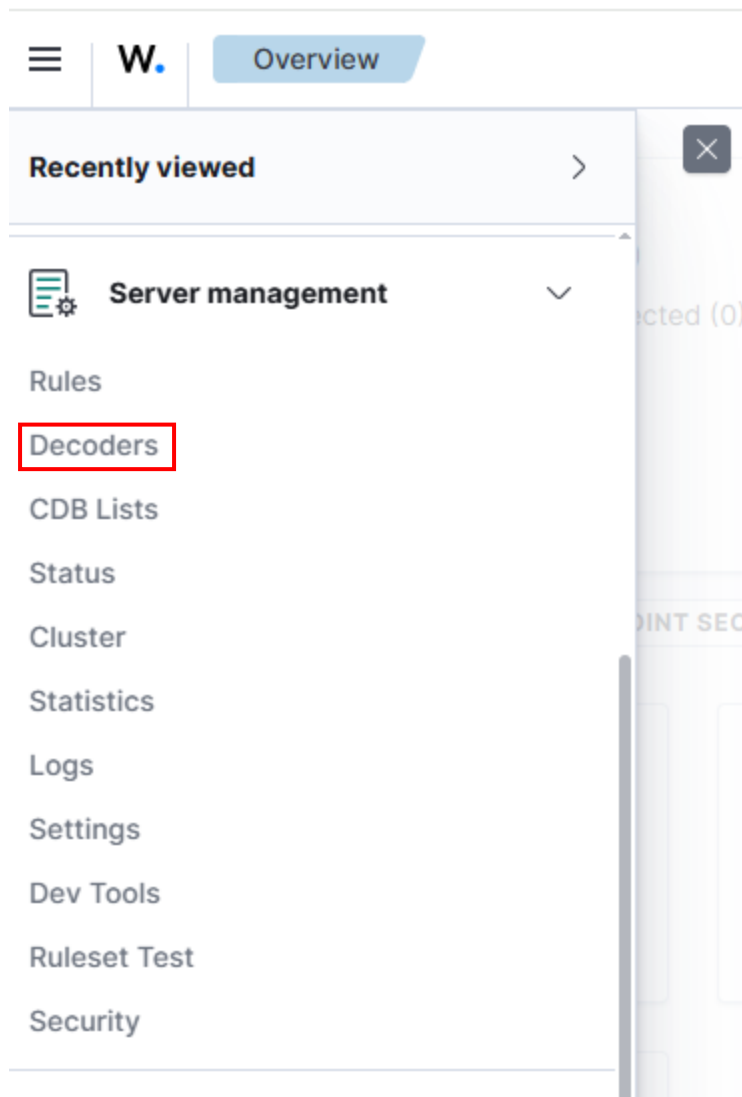
Target Categories

[Restricted_Sites]	access	deny
Default access [all]	access	deny

Do not allow IP-Addresses in URL ☒ To make sure that people do not bypass the URL filter by simply using the IP-Addresses instead of the FQDN you can check this option. This option has no effect on the whitelist.

5. Create Custom Wazuh Rules for SquidGuard

On Wazuh Manager Create new Decoder first:



On wazuh Manager create new rules for squidguard:

< SquidGuardRules.xml

Ruleset Test Save

```
1 <group name="squidguard,custom,">
2   <rule id="100904" level="5">
3     <decoded_as>squidguard-custom</decoded_as>
4     <field name="squidguard_action">DENIED</field>
5     <description>SquidGuard blocked website access.</description>
6     <group>web_filter,pci_dss_1.4,gpg13_4.12</group>
7   </rule>
8 </group>
9
```

6. Output of the following Rules and also check logs in pfsense and wazuh:



Sorry! Something went wrong :(

Last 500 Firewall Log Entries. (Maximum 500)						
Action	Time	Interface	Rule	Source	Destination	Protocol
✖	Jul 8 06:48:22	WAN	Block private networks from WAN block 192.168/16 (12004)	📡 192.168.100.84:137	📡 192.168.100.255:137	UDP
✖	Jul 8 06:48:23	WAN	Block private networks from WAN block 192.168/16 (12004)	📡 192.168.100.84:137	📡 192.168.100.255:137	UDP
✖	Jul 8 06:48:25	WAN	Block private networks from WAN block 192.168/16 (12004)	📡 192.168.100.84:137	📡 192.168.100.255:137	UDP
✖	Jul 8 06:48:25	WAN	Block private networks from WAN block 192.168/16 (12004)	📡 192.168.100.84:137	📡 192.168.100.255:137	UDP
✖	Jul 8 06:48:27	WAN	Block private networks from WAN block 192.168/16 (12004)	📡 192.168.100.84:137	📡 192.168.100.255:137	UDP
✖	Jul 8 06:48:28	WAN	Block private networks from WAN block 192.168/16 (12004)	📡 192.168.100.84:137	📡 192.168.100.255:137	UDP
✖	Jul 8 06:48:28	WAN	Block private networks from WAN block 192.168/16 (12004)	📡 192.168.100.84:137	📡 192.168.100.255:137	UDP
✖	Jul 8 06:48:29	WAN	Block private networks from WAN block 192.168/16 (12004)	📡 192.168.100.84:137	📡 192.168.100.255:137	UDP
✖	Jul 8 06:48:30	WAN	Block private networks from WAN block 192.168/16 (12004)	📡 192.168.100.84:137	📡 192.168.100.255:137	UDP
✖	Jul 8 06:48:31	WAN	Block private networks from WAN block 192.168/16 (12004)	📡 192.168.100.84:137	📡 192.168.100.255:137	UDP
✖	Jul 8 06:48:31	WAN	Block private networks from WAN block 192.168/16 (12004)	📡 192.168.100.84:137	📡 192.168.100.255:137	UDP
✖	Jul 8 06:48:32	WAN	Block private networks from WAN block 192.168/16 (12004)	📡 192.168.100.84:137	📡 192.168.100.255:137	UDP
✖	Jul 8 06:48:34	WAN	Block private networks from WAN block 192.168/16 (12004)	📡 192.168.100.84:137	📡 192.168.100.255:137	UDP

2,226 hits

Jul 8, 2025 @ 11:49:51.703 - Jul 9, 2025 @ 11:49:51.704

Export Formatted

700 available fields

Columns

Density

1 fields sorted

Full screen

timestamp	agent.name	syscheck.path	syscheck.event	rule.des...	rule.level	rule.id
 Jul 9, 2025 @ 11:49:47.678	win10	c:\program files (x86)\ossec-agent\queue\diff\file\e983...	modified	Integrity ch...	7	550
 Jul 9, 2025 @ 11:49:43.573	win10	c:\program files (x86)\ossec-agent\queue\diff\file\dc0e...	modified	Integrity ch...	7	550
 Jul 9, 2025 @ 11:49:41.985	win10	c:\program files (x86)\ossec-agent\queue\diff\file\d5e1...	modified	Integrity ch...	7	550
 Jul 9, 2025 @ 11:49:36.643	win10	c:\program files (x86)\ossec-agent\queue\diff\file\c905...	modified	Integrity ch...	7	550
 Jul 9, 2025 @ 11:49:29.491	win10	c:\program files (x86)\ossec-agent\queue\diff\file\c6fb0...	modified	Integrity ch...	7	550
 Jul 9, 2025 @ 11:49:28.316	win10	c:\program files (x86)\ossec-agent\queue\diff\file\c583...	modified	Integrity ch...	7	550
 Jul 9, 2025 @ 11:49:26.991	win10	c:\program files (x86)\ossec-agent\queue\diff\file\c262...	modified	Integrity ch...	7	550
 Jul 9, 2025 @ 11:49:25.590	win10	c:\program files (x86)\ossec-agent\queue\diff\file\bf3ac...	modified	Integrity ch...	7	550
 Jul 9, 2025 @ 11:49:25.032	win10	c:\program files (x86)\ossec-agent\queue\diff\file\bddb...	modified	Integrity ch...	7	550
 Jul 9, 2025 @ 11:49:24.602	win10	c:\program files (x86)\ossec-agent\queue\diff\file\bd0d...	modified	Integrity ch...	7	550
 Jul 9, 2025 @ 11:49:24.412	win10	c:\program files (x86)\ossec-agent\queue\diff\file\bd05...	modified	Integrity ch...	7	550

4. Administrator Privileges Rules

Configure Admin Access Rule:

Go to Firewall create rules on WAN:

Rules (Drag to Change Order)												
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions	
☒	0/270 KiB	*	RFC 1918 networks	*	*	*	*	*		Block private networks		
☒	0/4 KiB	*	Reserved Not assigned by IANA	*	*	*	*	*		Block bogon networks		
☐	0/0 B	IPv4 TCP	192.168.100.20	*	192.168.100.30	80 - 443	*	none				
☐	0/0 B	IPv4 *	pfB_Asia_v4	*	*	*	*	none		pfB_Asia_v4 auto rule		
☐	0/0 B	IPv6 *	pfB_Asia_v6	*	*	*	*	none		pfB_Asia_v6 auto rule		
☐	0/0 B	IPv4 *	pfB_Europe_v4	*	*	*	*	none		pfB_Europe_v4 auto rule		
☐	0/0 B	IPv4 TCP/UDP	*	*	Blockfacebook	80 - 443	*	none		Facebook Blocked on my site		
Add Add Delete Toggle Copy Save Separator												

Add Rules to Block all other access

Firewall / Rules / Edit

Edit Firewall Rule

Action

Block

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

WAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP

Choose which IP protocol this rule should match.

Source any and Destination port 80, 443, 22

Source

Source

☐ Invert match

Any

Source Address

/

Display Advanced

The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, **any**.

Destination

Destination

☐ Invert match

Address or Alias

192.168.1.1

/

Destination Port Range

HTTP (80)

From

Custom

To

HTTPS (443)

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log

☒ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset and displayed in the firewall log.

Test in Home Lab:

Jul 9 01:18:27	lighttpd_pfb	553	[pfBlockerNG] DNSBL Webserver started
Jul 9 01:18:28	tail_pfb	2231	[pfBlockerNG] Firewall Filter Service started
Jul 9 01:18:30	php_pfb	3173	[pfBlockerNG] filterlog daemon started
Jul 9 01:18:30	php	2507	[pfBlockerNG] DNSBL parser daemon started
Jul 9 01:27:04	php-fpm	569	/index.php: Successful login for user 'admin' from 192.168.2.1 (Local Database)
Jul 9 01:35:00	php-fpm	391	/pfblockerng/pfblockerng_Asia.php: Configuration Change: admin@192.168.2.1 (Local Database): [pfBlockerNG] save GeoIP [Asia] settings
Jul 9 01:35:00	check_reload_status	406	Syncing firewall
Jul 9 01:35:16	php-fpm	391	/pfblockerng/pfblockerng_update.php: Configuration Change: admin@192.168.2.1 (Local Database): Removed cron job for pfblockerng.php cron
Jul 9 01:35:16	check_reload_status	406	Syncing firewall

Check pfsense logs:

Status / System Logs / Firewall / Normal View

System

Firewall

DHCP

Authentication

IPsec

PPP

PPPoE/L2TP Server

OpenVPN

NTP

Packages

Settings

Normal View

Dynamic View

Summary View

Last 500 Firewall Log Entries. (Maximum 500)

Action	Time	Interface	Rule	Source	Destination	Protocol
✖	Jul 9 05:26:36	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.191:7551	192.168.100.255:7551	UDP
✖	Jul 9 05:26:38	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.73:137	192.168.100.255:137	UDP
✖	Jul 9 05:26:38	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.73:137	192.168.100.255:137	UDP
✖	Jul 9 05:26:38	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.191:7551	192.168.100.255:7551	UDP
✖	Jul 9 05:26:40	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.73:137	192.168.100.255:137	UDP
✖	Jul 9 05:26:40	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.100.191:7551	192.168.100.255:7551	UDP

Check in Wazuh Events

6,488 hits							
Jul 2, 2025 @ 15:55:57.256 - Jul 9, 2025 @ 15:55:57.256							
 Export Formatted  700 available fields  Columns  Density  1 fields sorted  Full screen							
	timestamp	agent.name	syscheck.path	syscheck.event	rule.des...	rule.level	rule.id
	Jul 9, 2025 @ 15:55:41.441	win10	c:\program files (x86)\ossec-agent\queue\syscollector\...	modified	Integrity ch...	7	550
	Jul 9, 2025 @ 15:55:41.387	win10	c:\program files (x86)\ossec-agent\queue\syscollector\...	modified	Integrity ch...	7	550
	Jul 9, 2025 @ 15:55:40.696	win10	c:\program files (x86)\ossec-agent\wazuh-logcollector....	modified	Integrity ch...	7	550
	Jul 9, 2025 @ 15:55:40.078	win10	c:\program files (x86)\ossec-agent\queue\logcollector\...	modified	Integrity ch...	7	550
	Jul 9, 2025 @ 15:55:39.998	win10	c:\program files (x86)\ossec-agent\wazuh-agent.state	modified	Integrity ch...	7	550
	Jul 9, 2025 @ 15:55:39.837	win10	c:\program files (x86)\ossec-agent\wazuh-agent.state	modified	Integrity ch...	7	550
	Jul 9, 2025 @ 15:54:05.518	win10	c:\program files (x86)\ossec-agent\wazuh-logcollector....	modified	Integrity ch...	7	550
	Jul 9, 2025 @ 15:54:04.890	win10	c:\program files (x86)\ossec-agent\queue\logcollector\...	modified	Integrity ch...	7	550
	Jul 9, 2025 @ 15:54:04.785	win10	c:\program files (x86)\ossec-agent\wazuh-agent.state	modified	Integrity ch...	7	550
	Jul 9, 2025 @ 15:50:26.295	win10	c:\program files (x86)\ossec-agent\wazuh-logcollector....	modified	Integrity ch...	7	550

Summery

- Implement firewall protection with pfsense
- Monitored firewall activity with wazuh
- Block Traffic by GeoIP and DNS
- Manage user Permission
- Generate traffic reports for security analysis.